

PH.D. QUALIFYING EXAM 2022: PROBABILITY

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “information sheet” for EACH OF THE TWO COURSES (Stat 592 and Stat 654) on which this exam is based. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. At the end of the exam, separate out the bunches of pages for problems 1 and 2, and hand them over to the proctor. Good luck!

Problem 1

Let $Y_1, \dots, Y_n$ be independent random variables such that, for each $i \geq 1$, Y_i is distributed according to a distribution having density

$$(0.1) \quad f_i(y) = ie^{-iy} \mathbf{1}_{\{y>0\}}$$

with respect to Lebesgue measure, for each $i = 1, \dots, n$. We define a random ordering $\prec_Y$ of the set $\{1, \dots, n\}$ from the order statistics of $\{Y_1, \dots, Y_n\}$ by putting

$$(0.2) \quad i \prec_Y j \iff Y_i < Y_j.$$

- (i) For each $i = 1, \dots, n$, compute the probability that $i \prec_Y j$ for all $j \neq i$.
(ii) Compute the probability that $\prec_Y$ corresponds to

$$i \prec_Y j \iff i < j,$$

where $<$ is the usual ordering of positive integers $1 < 2 < \dots < n$.

- (iii) Compute the probability that $\prec_Y$ corresponds to

$$i \prec_Y j \iff j < i,$$

where $<$ is the usual ordering of positive integers $1 < 2 < \dots < n$ as in (ii).

- (iv) Let P_n denote the probability from (iii) and Q_n denote the probability from (ii). Show that $(P_n/Q_n)^{1/n} \sim 4$ as $n \rightarrow \infty$, that is,

$$\lim_{n \rightarrow \infty} \frac{(P_n/Q_n)^{1/n}}{4} = 1.$$

Hint: You may use Stirling's approximation for factorial.

- (v) Suppose we first generate $\prec_Y$ as an ordering of $1, \dots, n$ based on $Y_1, \dots, Y_n$ with distribution in (0.1). For $1 < m < n$, we define $Y'_1, \dots, Y'_m$ by $Y'_1 = Y_1, \dots, Y'_m = Y_m$ and define an ordering $\prec_{Y'}$ of $1, \dots, m$ as in (0.2). Express the distribution of $\prec_{Y'}$ in terms of the distribution of $\prec_Y$.

Now consider the following sampling process. Let $Z_1, Z_2, \dots$ be independent random variables with distribution

$$\Pr(Z_i = j) \propto i - j, \quad j = 0, 1, \dots, i - 1.$$

For any n , construct a random ordering $\prec_{Z(n)}$ of $1, \dots, n$ from $Z_1, \dots, Z_n$ recursively by

- $\prec_{Z(1)}$ is the trivial ordering of 1 element with probability 1.
- Given $\prec_{Z(n)}$, define $\prec_{Z(n+1)}$ to be consistent with $\prec_{Z(n)}$ with element $n+1$ inserted into the ordering of $\prec_{Z(n)}$ in the $(Z_{n+1} + 1)$ position.

In particular, for $1 \leq i, j \leq n$, we have

$$i \prec_{Z(n+1)} j \quad \text{if and only if} \quad i \prec_{Z(n)} j,$$

and, writing

$$X_1 \prec_{Z(n)} X_2 \prec_{Z(n)} \cdots \prec_{Z(n)} X_n$$

to denote the random ordering determined by $Z^{(n)}$,

$$\begin{aligned} X_i &\prec_{Z(n+1)} (n+1), & \text{for all } i \leq Z_{n+1} \\ (n+1) &\prec_{Z(n+1)} X_i, & \text{for all } i > Z_{n+1}. \end{aligned}$$

For example, suppose $Z_1 = 0$, $Z_2 = 0$, $Z_3 = 1$, and $Z_4 = 3$. Then the resulting orderings for the first few values of n are

- $\prec_{Z(1)}: 1$
- $\prec_{Z(2)}: 2 \prec_{Z(2)} 1$
- $\prec_{Z(3)}: 2 \prec_{Z(3)} 3 \prec_{Z(3)} 1$
- $\prec_{Z(4)}: 2 \prec_{Z(4)} 3 \prec_{Z(4)} 1 \prec_{Z(4)} 4$

The above protocol gives rise to an infinite random ordering $Z^{(\infty)}$ on the positive integers defined by

$$i \prec_{Z^{(\infty)}} j \quad \text{if and only if} \quad i \prec_{Z^{(i \vee j)}} j,$$

where $i \vee j := \max\{i, j\}$.

For each $n = 1, 2, \dots$, let M_n be the highest ranked integer in $Z^{(n)}$, i.e., M_n is the random variable such that for every $i = 1, \dots, n$ either $M_n = i$ or $i \prec_{Z^{(n)}} M_n$. Define $M_\infty := \lim_{n \rightarrow \infty} M_n$.

(vi) Prove that M_∞ exists and is finite with probability 1.

Problem 2

Let n be a positive integer. Consider a simple random walk on $\mathbb{Z}^2$ starting at $(n, 0)$. At each step, the random walker moves one step towards one of the following directions: right $(1, 0)$, left $(-1, 0)$, up $(0, 1)$, down $(0, -1)$, each with probability $1/4$. Let $S_k = (S_{k,x}, S_{k,y}) \in \mathbb{Z}^2$ be the position of the walker at time k .

Part A:

- Is $\{S_{k,x}\}_{k \geq 0}$ a Markov chain? If it is, please specify its state space and transition probability distribution.
- Is $\{S_{k,x}\}_{k \geq 0}$ a martingale?

Part B: Let $T_{0,x}$ be the first time the walker hits the y -axis. That is, the first time the random walker's x -coordinate equals 0.

- Calculate $\mathbb{P}[T_{0,x} = \infty]$.
- Calculate $\mathbb{E}[T_{0,x}]$.

Part C: Let T_0 be the first time the walker arrives at the origin $(0, 0)$. In this problem, $\lfloor x \rfloor$ denotes the largest integer that is no larger than x . For example, $\lfloor 1.5 \rfloor = 1$, $\lfloor 4 \rfloor = 4$.

- Let D_k be the (Euclidean) distance between the random walker and the origin at time k . Let $P_{1,n} := \mathbb{P}[D_{\lfloor n^{1.5} \rfloor} \geq n/2]$ be the probability that the distance between the walker and the origin is at least $n/2$ after $\lfloor n^{1.5} \rfloor$ steps. Prove that $P_{1,n} \rightarrow 1$ as $n \rightarrow \infty$.
[Hint: There are more than one way to show this result. You can try to prove the complementary event's probability goes to 0. You can use the location of $S_{k,x}$ to upper bound the probability of the complementary event.]
- Let $P_{2,n} := \mathbb{P}[T_0 \leq \lfloor n^{1.5} \rfloor]$ be the probability that the walker has visited the origin in the first $\lfloor n^{1.5} \rfloor$ steps. Prove that $P_{2,n} \rightarrow 0$ as $n \rightarrow \infty$. [Hint: the maximal inequalities may be helpful. See below]

You may find the Doob's maximal inequality, or Kolmogorov's maximal inequality useful. These inequalities are given below.

- (Doob's maximal inequality) Let $\{X_k\}$ be a submartingale. Then

$$\mathbb{P}\left[\max_{1 \leq i \leq k} X_i \geq \lambda\right] \leq \frac{\mathbb{E}[\max\{X_k, 0\}]}{\lambda}.$$

- (Kolmogorov's Maximal Inequality) Let $\xi_1, \dots, \xi_k, \dots$ be a sequence of *i.i.d.* random variables with mean 0 and finite variance. Let $T_k := \sum_{i=1}^k \xi_i$. Then

$$\mathbb{P}[\max_{1 \leq i \leq k} |T_i| \geq \lambda] \leq \frac{\text{Var}[T_k]}{\lambda^2}.$$